Fall 2025-2026

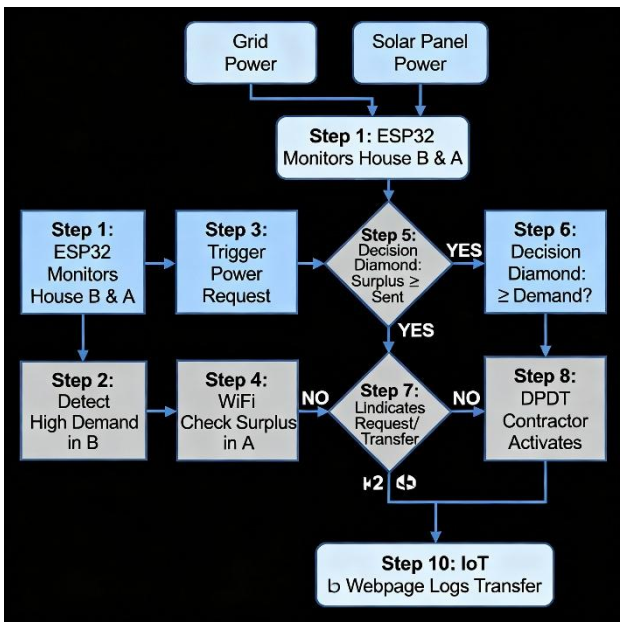
BEEE304L-Power Systems Engineering

SUSTAINABLE ENERGY EXCHANGE BETWEEN HOUSEHOLDS USING SOLAR POWER INTEGRATION

Objectives:

- To design a peer-to-peer (P2P) power-sharing system that enables House B to buy surplus solar energy from House A during high-demand periods.
- To develop a control strategy that manages power flow between House A, House B, and EB, ensuring stable and secure operations.
- To simulate and analyze the performance, reliability, and economic impacts of the proposed system under varying consumption and generation scenarios.
- To propose metering and billing mechanisms for fair accounting of energy exchanges between the houses.

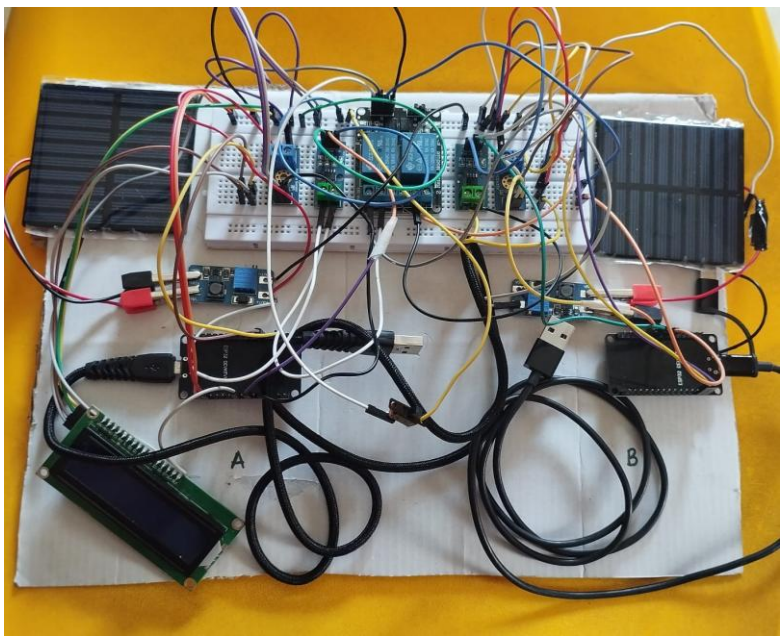
Block diagram:



Specifications:

- Microcontrollers (ESP32): Provide the core logic and handle webhosting for the system.
- Current/Voltage Sensors: Enable real-time monitoring of power parameters in the power flow.
- Relay Modules: Serve as the key switching mechanism for controlled power transfer, directly managed by the uCs.
- LCD Display & Web Interfaces

Hardware implementation:



Results and discussion:

- Successfully built a functional hardware prototype.
- The ESP32 control system performed reliably under all three test scenarios (P2P sharing, grid fallback, and standby).
- Real-time data logging via the IoT webpage successfully recorded all energy sharing behavior.

Conclusion:

This project validated a decentralized, peer-to-peer (P2P) energy-sharing framework [1], successfully integrating distributed solar generation for localized power trading[222]. The prototype confirms the technical viability of this architecture, proving its capability to optimize local energy utilization, reduce dependency on the central utility grid [3], and enhance local energy resilience[4].

References:

- Peer-to-Peer energy trading in a Microgrid Zhang, C., et al. (2018)
- Using peer-to-peer energy-trading platforms to incentivize prosumers to form federated power plantsMorstyn, T., et al. (2018) Using peer-to-peer energy-trading platforms to incentivize prosumers to form federated power plantsMorstyn, T., et al. (2018)

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